MILITARY SPECIFICATION

CONNECTORS, COAXIAL, RADIO FREQUENCY, SERIES LT,

GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the general requirements for RF, coaxial, weatherproof connectors (plugs, receptacles, and adapters) of the LT series having a nominal characteristic impedance of 50 ohms.

1.2 Classification. The connectors shall be of the type and part number covered as specified (see 3.1 and 6.2b).

1.2.1 Part number. The part number shall consist of the letter "M" followed by the basic specification sheet number, and a sequentially assigned dash number. The dash number (-1, -2, etc.) designates the material as brass and the finish as silver-plated in accordance with QQ-S-365, 0.0002 inch minimum.

Example:

M26637/1-1

Basic specification sheet number

Dash number from specification sheet (see 3.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-389	- Plastic Molding Material, FEP Flurocarbon, Molding and Extrusion.
O-F-499	- Flux, Brazing (Silver Brazing Filler Metal, Low Melting Point).
QQ-B-613	- Brass, Leaded and Nonleaded, Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-626	- Brass, Leaded and Nonleaded, Rod, Shapes, Forgings and Flat Product with Finished Edges (Bar and Strip).
QQ-B-654	- Brazing Alloy, Silver.
QQ-B-750	- Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-530	- Copper-Beryllium Alloy Bar, Rod and Wire (Copper Alloy Numbers 172 and 173).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: 2750th ABW/ES, Gentile Air Force Station, Ohio 45444-4500, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-C-533 - Copper-Beryllium Alloy Strip (Copper Alloy Numbers 170 and 172).
QQ-S-365 - Silver Plating, Electrodeposited, General Requirements for.
QQ-S-571 - Solder, Tin Alloy, Tin Alloy and Lead Alloy.
ZZ-R-765 - Rubber, Silicone.

MILITARY

MIL-F-14072 - Finishes for Ground Electronic Equipment.
MIL-I-17214 - Indicator, Permeability Low-Mu (Go-No-Go).
MIL-C-55330 - Connectors, Electrical and Fiber Optic, Packaging of.

(See supplement 1 for applicable specification sheets.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component parts.
MIL-STD-45662 - Calibration Systems Requirements.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Department of Defense

H4-1 - Federal Supply Code for Manufacturers.

National Bureau of Standards

H-28 - Screw Thread Standards for Federal Services.

(Copies of the specifications, standards and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1-1962 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1457 - Standard Specification for PTFE Molding and Extrusion Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Material. The connector body and coupling mechanism shall be brass with a silver finish. If materials other than those specified are used, the contractor shall certify to the user that the substitute material enables the connectors to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as guaranty of the acceptance of the product. When a definite material is not specified, a material shall be used which will enable the connector to meet the requirements of this specification. Materials shall be as specified herein (see table I).

TABLE I. Materials.

Component material	Applicable specification
Brass	QQ-B-613 or QQ-B-626
Copper beryllium	QQ-C-530 or QQ-C-533
Phosphor bronze	QQ-B-750
Flux	O-F-499
TFE fluorocarbon	ASTM D 1457
FEP fluorocarbon	L-P-389
Silicon rubber	ZZ-R-765
Silver solder	QQ-B-654
Soft solder	QQ-S-571

3.3.1 Finish. Unless otherwise specified (see 6.2), center and outer contacts shall be silver plated per QQ-S-365, .0002 inch minimum. All other metal parts will have a rhodium flash over the silver plating. The terminal ends of contacts shall be hot-tin-dipped so that the interior of the conductor holes (solder cups) shall have a coating of soft solder to facilitate assembly to the cable center conductor.

3.3.2 Dissimilar metals. Dissimilar metals between which an electromotive couple may exist shall not be placed in contact with each other. Reference is made to MIL-F-14072 for definition of dissimilar metals.

3.3.3 Nonmagnetic materials. All parts shall be made from materials which are classed as nonmagnetic (see 3.9).

3.4 Design and construction. Connectors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Mating (visual indication). When applicable (see 3.1), a visual means shall be provided to indicate when two mating connectors are properly mated.

3.5 Gauge tests. When connectors are tested as specified in 4.6.2, the center contacts and the other contacts shall meet the gauge tests as specified.

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3.6 Force to engage/disengage.

3.6.1 Threaded connectors. When tested as specified in 4.6.3.1, the torque necessary to completely couple or uncouple the connectors shall not exceed that specified (see 3.1).

3.7 Coupling proof torque. When tested as specified in 4.6.4, the coupling mechanism (threaded types) shall not be dislodged and the connector shall meet requirements of 3.6.1. The interface dimensions of the connector shall remain as specfied (see 3.1).

3.8 Mating characteristics. When connectors are tested as specified in 4.6.5, the mating dimensions shall be gauged as specified (see 3.1) and the dimensions shall remain within the specified tolerance (see 3.1).

3.9 Permeability of nonmagnetic materials. When connectors are tested as specified in 4.6.6, the permeability (mu) shall be less than 2.0.

3.10 Insulation resistance. When connectors are tested as specified in 4.6.7, the insulation resistance shall not be less than that specified (see 3.1).

3.11 Corrosion. When connectors are tested as specified in 4.6.8, there shall be no exposure of the base metal on the interface or mating surface, and they shall meet the requirements of 3.6.1, as applicable.

3.12 Voltage standing wave ratio (VSWR). When connectors are tested as specified in 4.6.9, the VSWR shall not exceed that specified over the frequency range specified (see 3.1).

3.13 Connector durability. When connectors are tested as specified in 4.6.10, they shall show no evidence of severe mechanical damage and the coupling device shall remain functional. Connectors shall meet the applicable requirements of 3.5 and 3.7.

3.14 Contact resistance. When connectors are tested as specified in 4.6.11, the contact resistance of the center contact, outer contact, and braid to connector shall be as specified (see 3.1).

3.15 Dielectric withstanding voltage. When connectors are tested as specified in 4.6.12, there shall be no evidence of breakdown.

3.16 Vibration. When the cabled (or wired, as applicable) connector is tested as specified in 4.6.13, there shall be no electrical interruptions exceeding 1 microsecond (µs), or as otherwise specified (see 3.1). There shall be no evidence of visual mechanical damage after the test, and the contact resistance of the center contact shall not be changed by more than the specified amount (see 3.1 and 3.14).

3.17 Shock (specified pulse). When the cabled (or wired, as applicable) connector is tested as specified in 4.6.14 there shall be no electrical interruptions exceeding 1 µs, unless otherwise specified (see 3.1). There shall be no evidence of visual or mechanical damage after the test, and the contact resistance of the center contact shall not be changed by more than the specified amount (see 3.1).

3.18 Thermal shock. After testing as specified in 4.6.15, there shall be no evidence of visual mechanical damage to the connector and it shall meet the dielectric withstanding voltage requirement (see 3.15), and the contact resistance specified for the center contact shall not be exceeded (see 3.14).

3.19 Altitude. When connectors are tested as specified in 4.6.16, they shall meet the corona and radio frequency high potential as specified (see 3.21 and 3.22).

3.20 Moisture resistance. When connectors are tested as specified in 4.6.17, there shall be no evidence of damage. They shall withstand the dielectric withstanding voltage specified (see 3.15), and the insulation resistance shall not be less than that specified (see 3.10).

3.21 Corona level. When connectors are tested as specified in 4.6.18, at the altitude and voltage specified (see 3.1), there shall be no evidence of sustained corona discharge.

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3.22 RF high potential withstanding voltage. When connectors are tested as specified in 4.6.19, there shall be no breakdown, or the leakage current specified shall not be exceeded (see 3.1).

3.23 Power handling capability. When connectors are tested as specified in 4.6.20, there shall be no evidence of material damage.

3.24 Cable retention force. When connectors are tested as specified in 4.6.21, there shall be no evidence of mechanical failure, loosening, rupture, or discontinuity.

3.25 Coupling mechanism retention force. When tested as specified in 4.6.22, the coupling mechanism shall not be dislodged from the connector and shall be capable of meeting the requirements of 3.6.1 immediately after the test.

3.26 Assembly instructions. Complete assembly instructions shall be furnished by the vendor with each connector procured under this specification. Assembly instructions shall include:

a. Cable preparation: Stripping dimensions and tolerances.

b. Pictorial presentation of sub-assemblies and loose piece parts.

c. Sufficient pertinent dimensions for verification of correct parts; as a minimum, the cable entry openings for conductor, dielectric, braid, and jacket shall be specified.

d. Recommended cable clamp tightening torque (if applicable).

e. Military part number and manufacturer's part number.

3.27 Marking. Connectors and associated fittings shall be permanently and legibly marked in accordance with the general marking requirements of MIL-STD-130 with the military part number (see 1.2.1) and the manufacturer's federal supply code. The marking location is optional; when practicable, a location should be picked that will least likely be covered in cable assembly or installation.

3.28 Workmanship. Connectors and associated fittings shall be processed in such a manner as to be uniform in quality and shall be free from sharp edges, burrs, and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202. For each test of threaded coupling connectors where the test is performed on mated pairs, the pair shall be torqued to the specified value (see 3.1).

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Twelve connectors of the same part number with its mating connector shall be subjected to first article inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into four groups consisting of three each. The sample units shall then be subjected to the inspection for their particular group.

4.4.3 Failures. Any failures shall be cause for refusal to grant lot approval.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all the connectors and associated fittings comprised of identical piece parts produced under essentially the same conditions and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III, and shall be made on the same set of sample units, in the order shown.

4.5.1.2.1 Sampling plan (group A inspection). Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.2 Rejected lots (group A inspection). If an inspection is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not, thereafter, be tendered for acceptance unless the former rejection or requirements of correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV in the order shown, and shall be made on sample units which have been subjected to and passed the group A inspection. Connectors having identical piece parts may be combined for lot purposes and shall be in proportion to the quantity of each part-numbered connector produced.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Method paragraph
Group I		
Visual and mechanical examination		
Material	3.3	
Finish	3.3.1	4.6.1
Dissimilar metals	3.3.2	
Design and construction (dimensions)	3.4	4.6.1.1
Marking	3.27	
Mating (visual indication)	3.4.1	4.6.1
Gauge tests	3.5	4.6.2
Force to engage/disengage threaded types	3.6.1	4.6.3.1
Coupling proof torque	3.7	4.6.4
Mating characteristics	3.8	4.6.5
Permeability of nonmagnetic materials	3.9	4.6.6
Workmanship	3.28	4.6.1
Insulation resistance	3.10	4.6.7
Group II		
Corrosion	3.11	4.6.8
Group III		
Voltage standing-wave ratio	3.12	4.6.9
Connector durability	3.13	4.6.10
Group IV		
Center contact resistance	3.14	4.6.11
Dielectric withstanding voltage	3.15	4.6.12
Vibration	3.16	4.6.13
Shock (specified pulse)	3.17	4.6.14
Thermal shock	3.18	4.6.15
Altitude	3.19	4.6.16
Moisture resistance	3.20	4.6.17
Corona level	3.21	4.6.18
RF high potential withstanding voltage	3.22	4.6.19
Power handling capability	3.23	4.6.20
Cable retention forces	3.24	4.6.21
Coupling mechanism retention force	3.25	4.6.22
Group V		
Contact resistance	3.14	4.6.11

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4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 2.5 percent defective.

4.5.1.3.2 Rejected lots (group B inspection). If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not, thereafter, be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination				
Material	3.3	4.6.1	.65	2.5
Finish 1/	3.3.1			
Dissimilar metals	3.3.2			
Design and construction	3.4			
Marking	3.27			
Workmanship	3.28			
Gauge tests	3.5			
Dielectric withstanding voltage	3.19	4.6.12		

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted. Any connector or connector part deformed or otherwise damaged during testing shall not be delivered on the contract or order.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Method paragraph
Force to engage/disengage		
Threaded type	3.6.1	4.6.3.1
Coupling proof torque	3.7	4.6.4
Mating characteristics	3.8	4.6.5
Permeability of nonmagnetic material	3.9	4.6.6
Insulation resistance	3.10	4.6.7
Voltage standing-wave ratio (uncabled)	3.12	4.6.9

4.5.2 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-55330.

4.5.3 Test conditions.

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4.5.3.1 Room ambient conditions. Unless otherwise specified, the test shall be conducted under the following conditions:

a. Temperature: Room ambient (13°C to 38°C).

b. Altitude: Normal ground.

c. Vibration: None.

d. Relative humidity: 20 to 80 percent.

4.5.3.2 Test condition tolerances. Unless otherwise specified, the maximum tolerance on test conditions shall be as follows:

a. Temperature: ±2°C.

b. Altitude: ±5 percent in feet.

c. Humidity: ±2 percent relative.

4.5.3.3 Stabilization of test conditions. When specified, temperature and altitude stabilization shall have been achieved prior to the start of the operational portion of the test in accordance with the following definitions:

a. Temperature stabilization. When the temperature of the chamber and the temperature of the test item do not change by more than ±2°C, nor does temperature of the chamber and the temperature of the test item differ by more than 3.5°C over a 10-minute period, the temperature shall be considered stabilized.

b. Altitude stabilization. When the altitude remains constant within ±5 percent of the specified value for 10 minutes, the altitude shall be considered stabilized.

4.5.4 Temperature sensing devices. When it is required during the test program to measure and record the temperature or associated temperature rise of the equipment, two thermocouples shall be used to measure ambient conditions and a minimum of three thermocouples shall be used on the test sample. The temperature of the thermocouples shall be recorded at the start and completion of the test period, at 5-minute intervals during the test period, and at more frequent intervals if the temperature rise exceeds 3°C during any 5-minute period.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. Connectors and associated fittings shall be inspected to verify that the design, construction, physical dimensions, assembly instructions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.27, and 3.28).

4.6.1.1 Dimensional inspection. Mating dimensions shall be inspected by mating the connector with its applicable mating gauges or other suitable means acceptable to the Government.

4.6.2 Gauge tests for contacts (see 3.5). Dimensions of gauges (pins and rings) used in the performance of these tests are based on the size of contacts being inspected and shall be as specified in the military standard. The pins may be tapered at their ends to facilitate insertion in the contacts, but the tapered portion shall not be included in the specified dimensions. All tests shall be performed in the order shown.

4.6.2.1 Center contacts (female). Female contacts shall be tested as follows:

a. All contact members shall make contact with pin A at least 1/32 inch from their tip ends when inserted in the sample.

b. Contact members shall not make contact with pin B when inserted in the sample.

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4.6.2.2 Outer contacts. All fingers of the slotted outer contact shall make contact within 1/32 inch of their tip ends, with the inside surface (inside diameter) of ring A and shall not make contact with the inside surface of ring B.

4.6.3 Force to engage/disengage.

4.6.3.1 Threaded connectors (see 3.6.1). The connector shall be engaged with its mating standard part (see 3.1). During the entire coupling/uncoupling cycle (until the connector is fully engaged/disengaged) the forces and/or torques necessary shall not exceed those specified (see 3.1). A thread coupled connector is fully engaged with its mating standard part when their reference planes (see 3.1) coincide. The mating standard part is a steel jig containing the critical interface dimensions finished to the tolerances specified (see 3.1). Its spring members, when applicable, shall be heat treated beryllium copper. The surface finish or mating surfaces shall be 16 microinches rms maximum, in accordance with ANSI B46.1-1962.

4.6.4 Coupling proof torque (see 3.7). The connector under test shall be engaged with its mating standard part (gauge) and the coupling nut tightened to the torque value specified (see 3.1). After 1 minute the connector under test and its mating standard part shall be disengaged.

4.6.5 Mating characteristics (see 3.8). After insertion of the specified oversize pin the specified number of times (see 3.1), the contact to be tested shall be held rigid by means of a suitable jig or fixture. A gauge, containing the test pin or test ring, and a suitable force indicating dial shall be aligned to within 0.004 TIR of any plane passing through the axis of the contact under test. Engagement or withdrawal of the test pin or test ring shall be made smoothly and at such a rate that the dial does not bounce or otherwise give a false reading. The test pin or test ring may be chamfered to facilitate entry, but the specified engagement length shall not include the chamfer length and the finish shall be as specified in accordance with ANSI B46.1-1962.

4.6.6 Permeability of nonmagnetic materials (see 3.9). The permeability of the connector shall be measured with an indicator conforming to MIL-I-17214.

4.6.7 Insulation resistance (see 3.10). Connectors without cables (when applicable) shall be tested in accordance with method 302, test condition B, MIL-STD-202. Measure between the center contact and body.

4.6.8 Corrosion (see 3.11). Unmated and uncabled connectors shall be tested in accordance with method 101, MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter (see 3.1).

b. Salt solution: 5 percent.

After exposure, connectors shall be washed, shaken, and lightly brushed as specified in method 101 of MIL-STD-202 and then permitted to dry for 24 hours at 40°C. Connectors shall then be examined for evidence of corrosion, pitting, and ease of coupling.

4.6.9 Voltage standing wave ratio (VSWR) (see 3.12). The VSWR shall be measured in accordance with the following procedure or a method acceptable to the Government. In the event of dispute, the method outlined herein shall be used. Diagrams for the swept frequency VSWR system check out and measurement procedures are shown on figure 1.

In the basic measurement setup of figure 1, detector 1 provides a feedback signal to the swept RF source in order to normalize the output signal of detector 2. The frequency-amplitude characteristics of detectors 1 and 2 should be matched within 0.5 dB.

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To measure VSWR, several sweeps are made with the slotted line probe incrementally positioned over at least a half wave length at the lowest frequency of interest. In this manner an X-Y display is generated whose upper and lower envelope limits represent maximum and minimum amplitudes of the standing wave for each frequency in the test band. A base line may be generated by making a sweep with no input to the measurement channel amplifier. The resultant X-Y display is calibrated according to the characteristics of the measurement channel detector and amplifier, e.g., linear, square law, logarithmic, etc.

The VSWR test system is checked out by successively terminating the slotted line with the elements shown in steps 1 and 2 and sweeping the frequency over the specified test band (see 3.1). In step 1, the system VWSR shall be less than 1.02 +.004 F (F measured in GHz). In step 2, the system VSWR shall be as specified (see 3.1).

For first article inspection (see table II), the system is checked out with the slotted line terminated as in step 3 using the specified cable (see 3.1). The impedance variation (random or periodic, or both) from the nominal characteristic impedance specified for the selected test cable should be no more than 1.0 percent when tested by time domain reflectometry. In step 3, the system VSWR shall be as specified (see 3.1). For first article inspection, the connector under test is measured with the slotted line terminated as in step 4. The VSWR shall be as specified (see 3.1).

Group B inspection tests (see table IV) are performed with the slotted line terminated as shown in step 7. The input part of the cable simulator must have the same interface configuration, dimensions, and dielectric as the recommended cable interface for the connector under test. The cable simulator for group B inspection tests must meet the specified VSWR (see 3.1) when tested as shown in step 6.

The standard precision adapter interface shall conform to IEEE Standard 287. Item 6 (standard precision adapters) shall not exceed the specified VSWR requirements (see 3.1). Standard test adapter designs shall be approved by the military qualifying agency.

4.6.10 Connector durability (see 3.13). Each connector under test shall be mated with a typical production connector in accordance with this specification. The connector shall be subjected to the number of cycles of mating and unmating specified (see 3.1). The connector and its mating part shall be completely engaged and completely disengaged during this cycle. Lubrication of the threads or rotational parts shall not be employed for this test unless specified (see 3.1). It is permissible to shake or blow debris from the threads or interface surfaces at intervals of not less than 50 cycles. Solvents or tools shall not be used for cleaning.

4.6.11 Contact resistance (see 3.14). All contact resistance tests shall be conducted with the apparatus shown on figure 2. Circuit adjustments and the measurement procedures for all contact resistance tests shall be in accordance with 4.6.11.1. The contact resistance to be measured are:

a. The contact resistance between the cable braid or outer conductor and the connector at the point of contact.

b. The contact resistance of the mated outer conductor contacts (the coupling nut must be removed for this measurement).

c. The contact resistance of the mated inner conductor contacts.

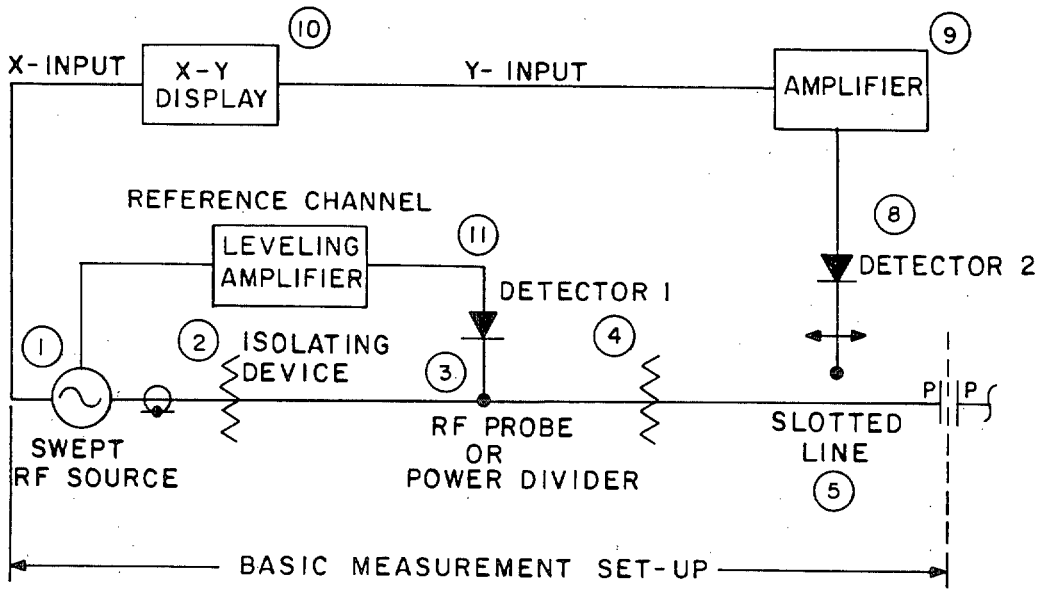
4.6.11.1 General procedures. The apparatus shall be assembled as shown in figure 2. The contacts, C_1 - C_2, shown in the figure represent the mating contacts upon which millivolt drop tests are to be conducted as follows:

a. Remove contacts C_1 - C_2 from the measuring circuit.

b. Close switch SW.

c. Adjust R_2 for a millivoltmeter (mVm) reading of 50 millivolts.

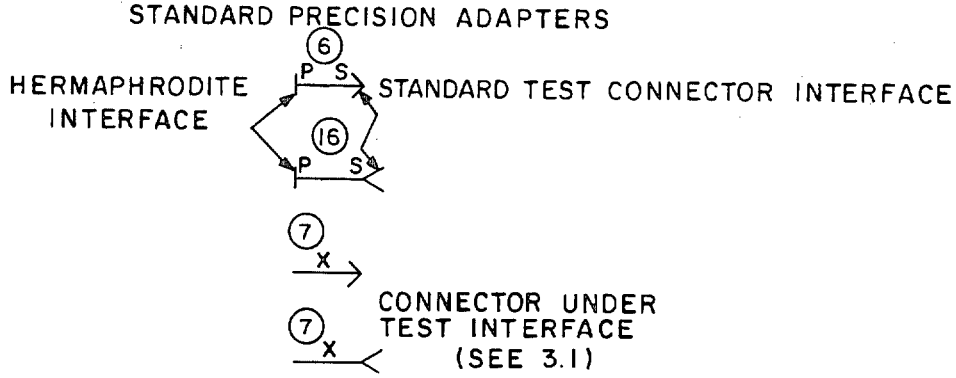
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MEASUREMENT CHANNEL

P P PRECISION HERMAPHRODITIC CONNECTORS

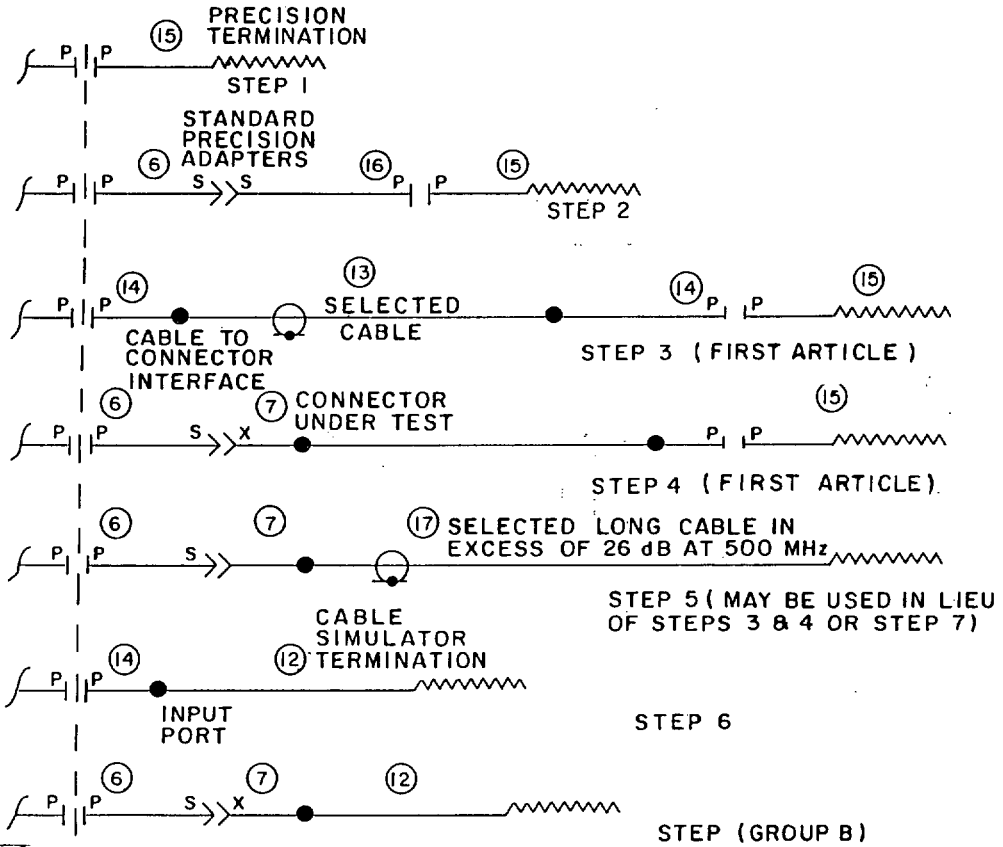
STANDARD PRECISION ADAPTERS



LEGEND

FIGURE 1. Swept frequency VSWR test.

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(1) Swept RF source.

(2) Isolating device.

(3) RF probe or power divider.

(4) Isolating device.

(5) Slotted line with precision hermaphroditic output connector. Residual VSWR less than 1.006 -.003F (F in GHz).

(6) Standard precision adapter maximum VSWR (see 3.1).

(7) Connector under test.

(8) Detector number 2.

(9) Amplifier.

(10) X-Y display.

(11) Detector number 1.

(12) Cable simulator terminiation.

(13) Selected test cable for first article.

(14) Precision hermaphroditic cable connector.

(15) Precision hermaphroditic termination.

(16) Standard precision adapters.

(17) Selected long cable whose attenuation at the lowest test frequency is 26 dB minimum (see 3.1).

FIGURE 1. Swept frequency VSWR test - Continued.

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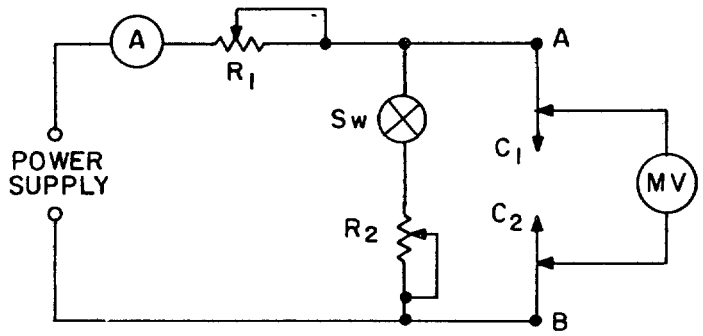


FIGURE 2. Diagram for contact resistance.

d. Connect contacts C_1 - C_2 to the measuring circuit and mate.

e. Check to see that mVm drops significantly prior to opening switch in (f).

f. Open switch SW.

g. Adjust R_1 for a circuit current (A) of 1 ampere.

h. Measure the millivolt drop across contacts C_1 - C_2 and call this "e".

i. Compute contact resistance. Contact resistance (milliohms) = e millivolts ÷ 1 ampere.

4.6.12 Dielectric withstanding voltage (see 3.15). Connectors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Special preparations or conditions.

(1) The maximum relative humidity shall be 50 percent. When facilities are not available at this test condition, connectors shall be tested at room ambient relative humidity. In case of dispute, if the test has been made at room ambient relative humidity, retest shall be made at 50 percent maximum relative humidity.

(2) The center contact of plug connectors and receptacle connectors shall be positioned in such a manner as to simulate actual assembly conditions.

(3) Precautions shall be taken to prevent air-gap voltage breakdowns.

(4) The voltage shall be metered on the high side of the transformer.

b. Magnitude of test voltage (see 3.1). The voltage shall be instantaneously applied.

c. Nature of potential: Alternating current.

d. Points of application of test voltage: Between the center contact and body.

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4.6.13 Vibration (see 3.16). A complete connector assembly shall be mounted as shown on figure 3 and vibrated in accordance with test condition B, method 204, MIL-STD-202. The center and outer contacts shall be connected to a suitable monitoring device. Suitable coaxial cable or wire as applicable, using the normal connecting devices of the connector and clamped as shown in figure 3, shall be used. At least 100 milliamperes shall be flowing through each set of contacts. Contacts may be connected in series. The connector shall be mounted by its normal mounting device and engaged by its normal coupling device. No safety wire shall be used. Cable to cable connectors may be held to the jig of figure 3 by a suitable clamp on one half of the connector assembly. The following conditions shall apply:

a. Test condition letter (see 3.1).

b. Continuity shall be monitored during vibration with a detector capable of detecting interruptions of 1 µs duration or longer, or as specified at 100 milliamperes.

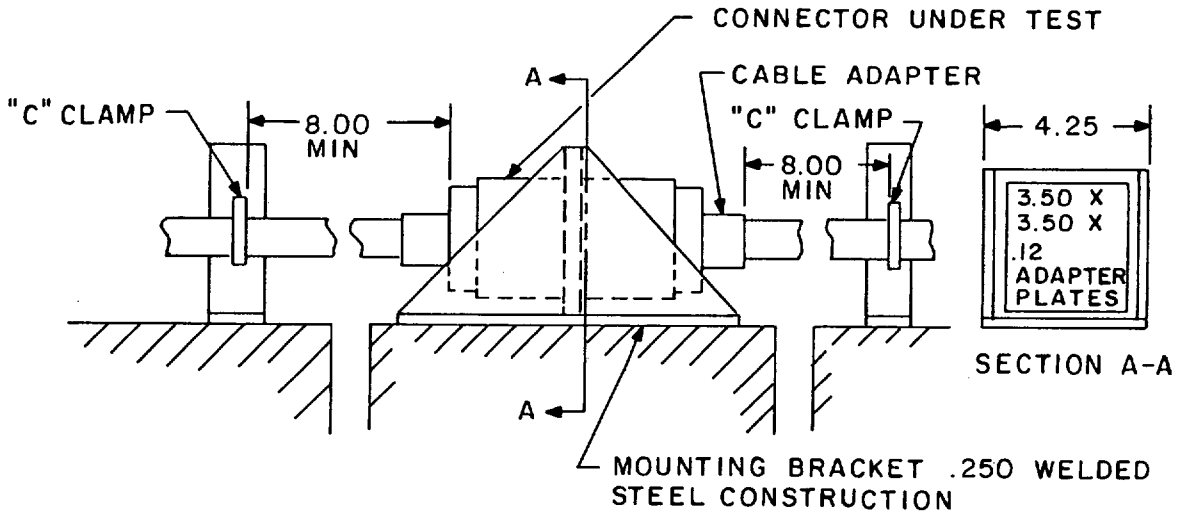


FIGURE 3. Vibration testing setup.

4.6.14 Shock (specified pulse) (see 3.17). The connector shall be mated with its mating connector (see 3.1) and subjected to method 213 of MIL-STD-202. The following exceptions and details shall apply:

a. Receptacles and panel or bulkhead mounted connectors and adapters shall be mounted by normal means. All other connectors and adapters shall be rigidly clamped to the vibrator table.

b. Acceleration requirements (see 3.1).

c. Three blows in each of three mutually perpendicular planes; one of which shall be parallel to the axis of the connector.

d. Continuity shall be monitored during shock as specified in 4.6.15(b).

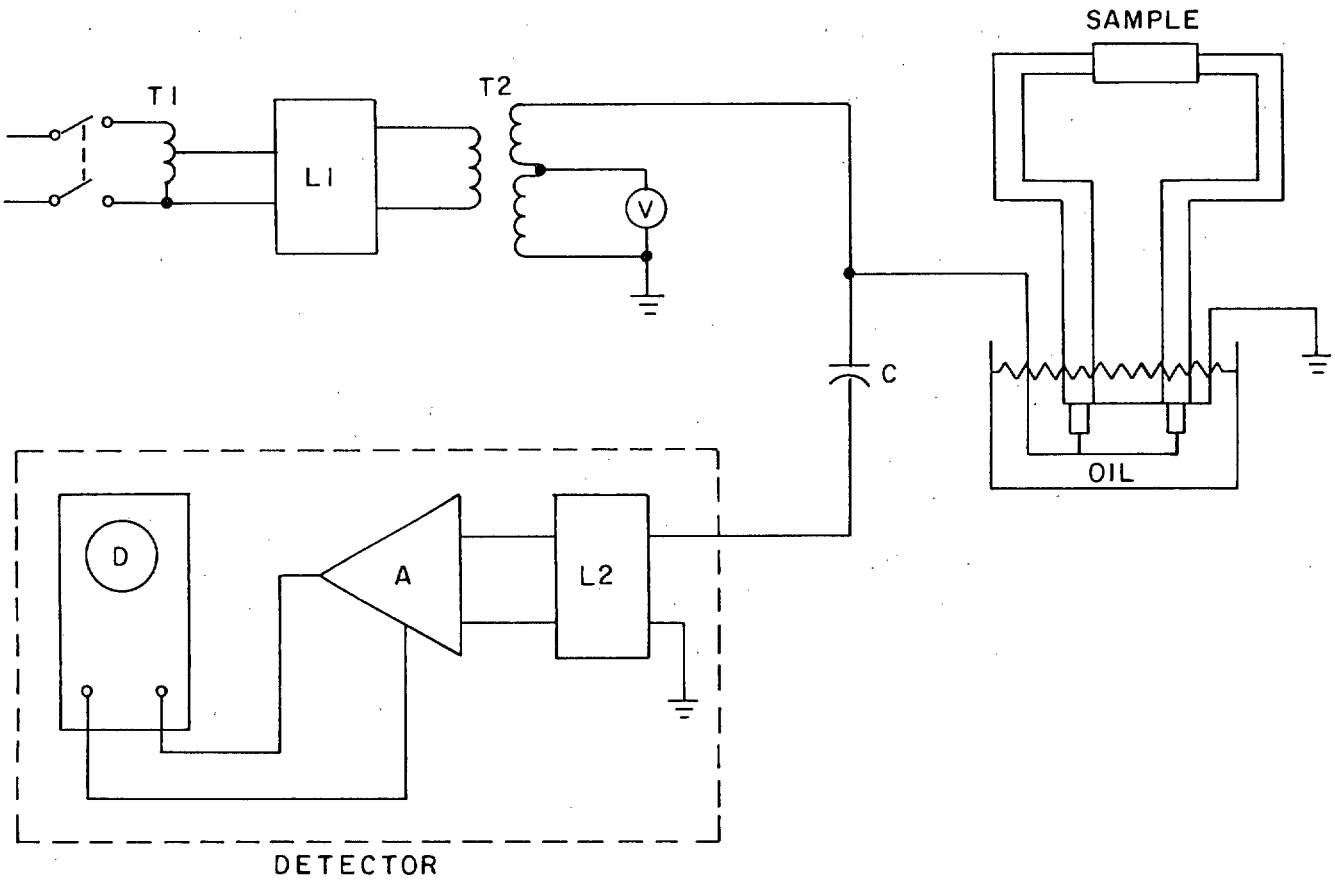
e. Center contact resistance shall be measured in accordance with 4.6.11 after the shock test.

4.6.15 Thermal shock (see 3.18). Connectors shall be subjected to method 107 of MIL-STD-202. The following details shall apply:

a. Test condition letter (see 3.1).

b. The contact resistance tests on the center contact shall be performed before and after the thermal shock test and examined for mechanical damage (see 3.1).

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C - Corona free coupling capacitor (note 1)
D - Discharge display
L1 - Input line filter (note 2)
L2 - 10-50 kHz detector input filter
A - Detector amplifier
T1 - 0-130 V variable transformer
T2 - High voltage transformer (corona free - less than 5 picocoulombs)
V - Voltmeter

NOTES:
1. Equal to or greater than total circuit capacitance.
2. 100 dB 14 kHz to 10 kMHz.

FIGURE 4. Equipment and schematic for measuring corona level.

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4.6.16 Altitude (see 3.19). Samples shall be subjected to the barometric pressure test of MIL-STD-202, method 105A, test condition C (70,000 feet) during which they shall be given the corona and the high potential tests herein. The applicable power shall be as specified (see 3.1).

4.6.17 Moisture resistance (see 3.20). The connector shall be mated and cabled with its mating connector and shall be subjected to method 106, MIL-STD-202. The following exceptions and conditions shall apply:

a. No initial measurements.

b. No load.

c. Step 7b (vibration) shall be omitted.

d. Measurements shall be made at high humidity when specified (see 3.1).

e. The connector shall withstand the dielectric withstanding voltage specified (see 4.6.12) after the drying period.

4.6.18 Corona level (see 3.21). The test sample shall be connected to a mating connector and arranged in a suitable test circuit such as indicated in figure 4. Components of the test circuit shall be corona free to the extent that a discharge of 5 pico coulombs or less can be measured when the 60 Hz test potential is increased to the value specified at the reduced pressure specified (see 3.1). The type of cable and length of cable used shall be as specified (see 3.1). No grease or similar compounds shall be used in or on the test item. After the sample is purged of air, the 60 Hz voltage shall be slowly increased until the detector, operated at sensitivity of 5 pico coulombs, indicates a sustained corona discharge. The voltage shall then be decreased until corona is at the 5 pico coulombs level or less. The latter value is being the corona level of the connector under test. The contractor may, at his own option, use a corona detector (which has been approved by the Government) for performing the test in lieu of the test set up of figure 4.

4.6.19 RF high potential withstanding voltage (see 3.22). Connectors shall be mated with their mating connectors (see 3.1) and approximately 2 inches of their standard cable (see 3.1) appropriately attached. This assembly shall then be inserted into the high impedance circuit as shown in figure 5, or equivalent, and instantaneously subjected to the RF voltage and frequency specified (see 3.1) between the center contact and body of the connectors. The duration of the test shall be 1 minute. The RF voltage source shall be frequency stabilized and have an approximate pure sine wave output with minimum harmonic content. Means shall be provided to indicate disruptive discharge and leakage current. The maximum leakage current shall be as specified (see 3.1).

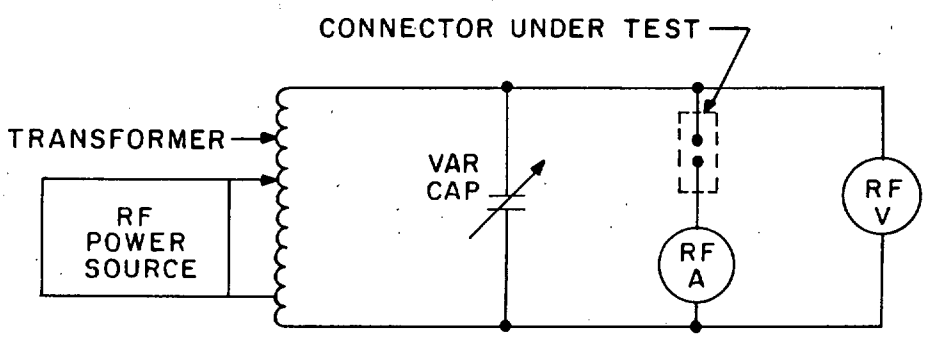


FIGURE 5. Circuit diagram for RF high potential withstanding voltage.

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4.6.20 Power handling capability test (see 3.23). The power handling capability of the test sample shall be determined by subjecting the assembly to rated power at the altitudes and temperatures specified herein.

4.6.20.1 Test setup.

4.6.20.1.1 Test equipment. A temperature-altitude chamber, high power terminations, a power source, and associated instrumentation shall be required.

4.6.20.1.2 Test conditions.

a. Temperature sensing devices (see 4.5.4) shall be used to measure chamber and test sample conditions, and chamber and test sample conditions shall be stabilized (see 4.5.3.3) prior to each step in the test program.

b. The upper frequency used shall be within 10 percent of the upper frequency limit of the sample under test. The lower frequency shall be within 0.10 GHz of the low end of the band, and the third frequency shall be the approximate center of the frequency range.

c. If a peak power rating is specified in the detail specification, the assembly shall be subjected to the peak power rating for 1 hour at a frequency within 10 percent of the high end of the band.

d. Test shall be conducted while the test sample is being subjected to the high temperature and high altitude conditions specified in 3.19.

e. The VSWR of the load shall be a minimum of 1.75:1.

4.6.20.2 Procedures.

a. The assembly shall be placed in the test chamber and the power handling capability of the test sample shall be determined by energizing the test sample at full rated power as specified in the detail specification.

b. The time period for the low and middle frequencies tested shall not be less than 20 minutes. The time period for the high frequency tested shall not be less than 1 hour.

c. The chamber and equipment temperature shall be returned to, and stabilized at, room temperature (4.5.3.1) between each application of high power.

d. The assembly shall be subjected to the peak power specified in the applicable detail specification for a period of 1 hour within 10 percent of the specified high frequency.

e. At the conclusion of the power handling test, the assembly shall be removed from the chamber and shall be subjected to examination and a gauge test as specified in 4.6.2.

4.6.20.3 Accept-reject criteria. If the assembly completes the power test without breakdown, the equipment will be considered to pass the test.

4.6.21 Cable retention force (see 3.24). When applicable (see 3.1), the connector shall be assembled to its standard mating test cable. The connector shall be firmly fixed and a movable sleeve attached to the cable. The sleeve is then moved longitudinally away from the fixed connector gradually and in such a manner that the cable remains unbent and untwisted. A scale for measuring the retention force (see 3.1) shall be attached to the sleeve. The force shall be held for 30 seconds minimum. The assembly shall then be examined for mechanical failure, loosening, or rupture and tested for continuity with a simple 115 volt, 60 Hz ac lamp circuit. With the connector still in the fixed position, the cable shall be held at a point ten times the diameter of the cable from the connector and a torque shall be applied in both directions as specified (see 3.1). The cable shall then be bent at a radius of 10 times the diameter of the cable starting at the connector at an angle of 90° ±5° from the axis of the connector, then reversed 180° ±10°. Repeat this procedure four times, then retest and reexamine as outlined above.

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4.6.22 Coupling mechanism retention force (see 3.25). The connector body and coupling mechanism shall be respectively secured to the lower and upper jaws of a tensile tester in an appropriate manner. A tensile load shall be applied at a rate of approximately 100 pounds per minute up to the force as specified and held at that value for 1 minute (see 3.1). During the 1 minute of steadily applied force, the coupling mechanism shall be rotated with respect to the connector body, two full revolutions in each direction.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330.

6. NOTES

6.1 Intended use. Connectors covered by this specification are intended for use in airborne RF applications from 30 to 5000 MHz and are designed for use with large size cables as specified (see 3.1).

6.2 Ordering data. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable detail specification.

c. The complete part number of the connector or fitting ordered.

d. Specific finish when required (see 3.3.1).

6.3 First article. When a first article inspection is required, the item shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be preproduction sample, a first production item, a sample selected from the first lot of production items, or it may be a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one, two, etc. unit(s). The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, test approval of the documents' first article.

6.4 Cross-reference of part numbers. For the substitutability relationship of items covered by this specification and items covered by superseded documents, see table V. However, all connectors in stock may be considered interchangeable with the new part number for a period of 1 year from the effective date of this specification.

TABLE V. Cross reference.

Superseded part number	Superseding part number
MS24281-1	M26637/1-1
MS24282-1	M26637/2-1
MS24283-1	M26637/3-1

6.5 Engineering information. Illustrations and additional engineering data on the connectors and fittings covered by this specification (see 3.1) are available in MIL-HBK-216, RF Transmission Lines and Fittings, copies of which are available upon request from the Naval Aviation Depot, Philadelphia, Pennsylvania.

6.5.1 Engineering parameters. The parameters of nominal impedance, voltage rating frequency range, and temperature range will be as specified.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-26637A(USAF)	Military Specification Connectors, Coaxial, Radio Frequency, Series LT, General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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